

HYDROGENATED NBR SYNTHESIS AND LOW-MOLECULAR-WEIGHT HNBR PROCESS LANDSCAPE

1. ABSTRACT

This report synthesizes patent evidence on hydrogenated nitrile rubber (HNBR) preparation, with emphasis on routes that control residual unsaturation, molecular weight, and processability. The target material is HNBR derived from nitrile rubber (NBR), typically a copolymer of a conjugated diene and an α,β -unsaturated nitrile, optionally containing additional comonomers. Across the selected patents, two major manufacturing themes emerge. The first is direct hydrogenation of NBR, including reactor/process innovations such as spinning disk reactors and homogeneous hydrogenation catalysts, aimed at lowering hydrogenation severity, reducing equipment cost, and improving throughput. The second is a one-step or tandem metathesis/hydrogenation strategy that simultaneously cleaves unsaturation and saturates the polymer backbone to produce low-molecular-weight HNBR with narrower molecular weight distributions and low residual double bond (RDB) content.

The landscape shows that conventional batch hydrogenation of NBR/HNBR is constrained by mass-transfer limitations, high hydrogen pressures, and viscosity increase during hydrogenation. Patents addressing this problem disclose hydrogen pressures from 3.5 to 100 bar in spinning disk reactors, and in prior art discussions note typical batch pressures of 65 to 90 bar. Other disclosures focus on catalyst selection, including rhodium- and ruthenium-based homogeneous systems, Wilkinson-type catalysts, and metathesis-active Ru/osmium complexes used in the presence of hydrogen and optional co-olefins. A parallel set of patents addresses downstream purification, especially removal of ruthenium residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins.

Formulation and application trends also appear: low-Mooney HNBR grades are targeted for seals, hoses, bearing pads, and flexible pipes, while peroxide-vulcanized HNBR compounds with controlled scorch time are used for bonding to PVDF liners in aggressive oilfield service. Overall, the evidence supports synthesis routes that combine controlled precursor composition, selective hydrogenation or simultaneous metathesis/hydrogenation, and post-treatment purification to achieve low-RDB, low-Mooney HNBR suitable for demanding elastomer applications.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US7666950B2
- Patent Title: Process for preparing hydrogenated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparing hydrogenated nitrile rubber by subjecting nitrile rubber to hydrogenation in a spinning disk reactor. The process is presented as a lower-pressure, lower-cost alternative to conventional batch hydrogenation, and it may be carried out with homogeneous hydrogenation catalysts, including rhodium- or ruthenium-based systems. - Hydrogen pressure: 3.5 bar to 100 bar — Claim 1 - Hydrogen pressure: 65 to 90 bar — Background discussion of batch hydrogenation - Mooney viscosity (NBR): 30 to 90 ML 1+4 @100°C — General description - Mooney viscosity (NBR): 30 to 70 ML 1+4 @100°C — General description - Mooney viscosity (NBR): 30 to 50 ML 1+4 @100°C — General description - Molecular weight Mw (NBR): 200,000 to 500,000 — General description - Molecular weight Mw (NBR): 200,000 to 400,000 — General description - Molecular weight Mn (NBR): 80,000 to 150,000 — General description - Polydispersity index PDI (NBR): 2.0 to 6.0 — General description - Polydispersity index PDI (NBR): 2.0 to 4.0 — General description - Mooney viscosity (HNBR): 55 to 120 ML 1+4 @100°C — Background discussion - Molecular weight Mw (HNBR): 200,000 to 500,000 — Background discussion - Residual double bond (RDB) content: up to 18% — Background discussion - Mooney viscosity (low Mooney HNBR): 2 to 50 ML 1+4 @100°C — Prior-art discussion - Molecular weight Mw (low Mooney HNBR): 30,000 to 250,000 — Prior-art discussion - Polydispersity index: typically less than 3 — Prior-art discussion - Conjugated diene content: 40 to 90 wt% — General description - Conjugated diene content: 60 to 85 wt% — General description - α,β -unsaturated nitrile content: 10 to 60 wt% — General description - α,β -unsaturated nitrile content: 15 to 40 wt% — General description - Additional comonomers: 0 to 40 wt% — General description - Additional comonomers: 0.1 to 40 wt% — General description - Additional comonomers: 1 to 30 wt% — General description

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is directly relevant because it addresses the core HNBR hydrogenation step and proposes a reactor configuration intended to improve hydrogen transfer and reduce process severity. It also preserves the standard NBR-to-HNBR precursor relationship while broadening catalyst options for selective hydrogenation.

Patent 2

- Patent Number: US7951875B2

- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR prepared by olefin metathesis of nitrile rubber followed by hydrogenation, with the metathesis/hydrogenation concept used to reduce molecular weight and narrow molecular weight distribution. The disclosure emphasizes metathesis-derived chain scission as a route to lower-Mooney HNBR and discusses ruthenium-based metathesis catalysts in the context of hydrogenated products. - Molecular weight MW: 30,000 to 250,000 g/mol — Claim 1 - Mooney viscosity: less than 10 ML 1+4@100°C — Claim 1 - Polydispersity index: less than 2.5 — Claim 1 - Molecular weight MW: 40,000 to 220,000 — Claim 2 - Polydispersity index: less than 2.3 — Claim 3 - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 15 wt.-% solid — Example 2 - Cement concentration: 6 wt.-% solid — Example 3 - Cement concentration: 6 wt.-% solid — Example 4 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Summary text - Mw: 54 kg/mol — Summary text - PDI: 2.0 — Summary text - Mw: 296 kg/mol — Starting polymer comparison - Mn: 85 kg/mol — Starting polymer comparison - PDI: 3.4 — Starting polymer comparison - Mw: 189,000 — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mw: 136,000 — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mw: 59,000 — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mw: 111,000 — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: ethylene-assisted metathesis/hydrogenation at 80°C with 0.05 phr catalyst gave reduced molecular weight and lower RDB over 240 min. - Example 2: increasing catalyst loading to 0.10 phr at the same temperature produced a different hydrogenation profile and lower Mw than the starting polymer. - Example 3: 0.25 phr catalyst at 6 wt.-% solids gave the lowest Mw and Mooney viscosity among the listed experiments. - Example 4: 1-hexene was used as co-olefin, showing that alternative olefin partners can also support molecular-weight reduction and hydrogenation. - Example 1 summary: HNBR 1 showed faster RDB reduction and lower PDI than the industrial benchmark HNBR 3.

Technical Relevance This patent is highly relevant because it targets low-molecular-weight HNBR, a key objective for improved processability while retaining HNBR chemistry. It provides explicit evidence that simultaneous metathesis/hydrogenation can reduce Mw, Mooney viscosity, and RDB in a single step.

Patent 3

- Patent Number: US7772328B2
- Patent Title: Low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses low-molecular-weight HNBR produced by olefin metathesis of nitrile rubber followed by hydrogenation, with the goal of obtaining lower Mooney viscosity and narrower molecular weight distribution than conventional HNBR. The process is described as a route to shaped articles and emphasizes ruthenium-based metathesis chemistry combined with hydrogenation. - Molecular weight MW: 30,000 to 250,000 g/mol — Claim 1 - Mooney viscosity: less than 10 ML 1+4@100°C — Claim 1 - Polydispersity index: less than 2.5 — Claim 1 - Molecular weight Mw: 40,000 to 220,000 — Claim 2 - Polydispersity index: less than 2.3 — Claim 3 - Cement concentration: 15 wt.-% solid — Example 1 - Cement concentration: 15 wt.-% solid — Example 2 - Cement concentration: 6 wt.-% solid — Example 3 - Cement concentration: 6 wt.-% solid — Example 4 - Ethylene pressure: 6.9×10^5 Pa — Example 1 - Ethylene pressure: 6.9×10^5 Pa — Example 2 - Ethylene pressure: 6.9×10^5 Pa — Example 3 - 1-hexene amount: 150 g — Example 4 - Reactor temperature: 80°C — Table 1 - Catalyst load: 0.05 phr — Table 1 - Catalyst load: 0.10 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Catalyst load: 0.25 phr — Table 1 - Mn: 27 kg/mol — Summary text - Mw: 54 kg/mol — Summary text - PDI: 2.0 — Summary text - Mw: 296 kg/mol — Starting polymer comparison - Mn: 85 kg/mol — Starting polymer comparison - PDI: 3.4 — Starting polymer comparison - Mw: 189,000 — Experiment 1 - PDI: 2.59 — Experiment 1 - Mooney viscosity: 43 — Experiment 1 - Mw: 136,000 — Experiment 2 - PDI: 2.27 — Experiment 2 - Mooney viscosity: 28 — Experiment 2 - Mw: 59,000 — Experiment 3 - PDI: 1.90 — Experiment 3 - Mooney viscosity: 3 — Experiment 3 - Mw: 111,000 — Experiment 4 - PDI: 2.02 — Experiment 4 - Mooney viscosity: 31 — Experiment 4

Relevant Experimental Evidence - Example 1: ethylene co-olefin with 0.05 phr catalyst at 80°C reduced RDB from 100 to 0.4 over 240 min. - Example 2: higher catalyst loading of 0.10 phr gave a different hydrogenation trajectory and lower Mw than the starting substrate. - Example 3: 0.25 phr catalyst produced the lowest Mooney viscosity and Mw among the listed runs. - Example 4: 1-hexene co-olefin demonstrated that non-ethylene olefins can also be used in the metathesis/hydrogenation sequence. - Table 2: the metathesized product reached PDI 1.90 and Mooney viscosity 3, showing strong molecular-weight reduction.

Technical Relevance This patent is relevant because it demonstrates a route to low-Mooney HNBR through tandem metathesis and hydrogenation, which is directly aligned with the target of low residual unsaturation and improved processability. It also provides comparative evidence that

catalyst choice strongly affects hydrogenation efficiency and final polymer properties.

Patent 4

- Patent Number: EP2473281B1
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a one-step process for producing low-molecular-weight HNBR by simultaneously subjecting nitrile rubber to metathesis and hydrogenation in the presence of hydrogen and a ruthenium- or osmium-based catalyst. The process is explicitly positioned as an alternative to sequential metathesis followed by hydrogenation and is intended to deliver low-RDB HNBR with reduced molecular weight and narrower distribution. - Molecular weight Mw: 20,000 to 250,000 — Description of inventive process - Mooney viscosity: 1 to 50 ML 1+4 @100°C — Description of inventive process - Polydispersity index: less than 3.0 — Description of inventive process - Conjugated diene: C4-C6 — General description - Conjugated diene: 1,3-butadiene — General description - Conjugated diene: isoprene — General description - Conjugated diene: 2,3-dimethylbutadiene — General description - Conjugated diene: piperylene — General description - α,β -unsaturated nitrile: C3-C5 — General description - α,β -unsaturated nitrile: acrylonitrile — General description - α,β -unsaturated nitrile: methacrylonitrile — General description - α,β -unsaturated nitrile: ethacrylonitrile — General description - Unsaturated monocarboxylic/dicarboxylic acids: fumaric acid, maleic acid, acrylic acid, methacrylic acid — General description - Alkyl esters of α,β -unsaturated carboxylic acids: methyl acrylate, ethyl acrylate, butyl acrylate, butyl methacrylate, 2-ethylhexyl acrylate, 2-ethylhexyl methacrylate, octyl acrylate — General description - NBR Mooney viscosity: 24 to 70 ML 1+4 at 100°C — General description - NBR Mooney viscosity: 28 to 40 ML 1+4 at 100°C — General description - NBR Mw: 200,000 to 400,000 — General description - NBR Mw: 200,000 to 500,000 — General description - NBR composition: 40 to 85 wt% diene — Claim 5 - NBR composition: 14 to 50 wt% unsaturated nitrile — Claim 5 - NBR composition: 1 to 10 wt% additional monomers — Claim 5

Relevant Experimental Evidence - Example a: the patent cites prior metathesis catalysts and explains why they are not necessarily suitable for nitrile rubber degradation and hydrogenation. - Example a: the disclosure contrasts sequential metathesis/hydrogenation with the claimed simultaneous process in hydrogen. - Example a: the patent identifies catalyst families based on Hoveyda and Grela-type Ru complexes as suitable for the inventive process. - Example a: the target product is described as low-RDB HNBR with Mw 20,000 to 250,000 and Mooney 1 to 50. - Example a: the process is framed as a single-step route to lower molecular weight and narrower MWD.

Technical Relevance This patent is one of the most directly relevant disclosures because it claims simultaneous hydrogenation and metathesis of NBR to make low-molecular-weight HNBR. It provides the clearest process-level evidence for controlling Mw, Mooney viscosity, and residual unsaturation in a single step.

Patent 5

- Patent Number: US9150669B2
- Patent Title: Process for the preparation of low molecular weight hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a process for producing low-molecular-weight HNBR by reacting nitrile rubber in the presence of hydrogen, optional co-olefin, and a ruthenium- or osmium-containing catalyst of formula (I). The process is explicitly described as simultaneous hydrogenation and metathesis, intended to generate HNBR with low residual double bond content and narrower molecular weight distribution. - Molecular weight Mw: 20,000 to 250,000 — Description of inventive process - Mooney viscosity: 1 to 50 ML 1+4@100°C — Description of inventive process - Polydispersity index: less than 3.0 — Description of inventive process - Conjugated diene: C4-C6 — General description - Conjugated diene: 1,3-butadiene — General description - Conjugated diene: isoprene — General description - Conjugated diene: 2,3-dimethylbutadiene — General description - Conjugated diene: piperylene — General description - α,β -unsaturated nitrile: C3-C5 — General description - α,β -unsaturated nitrile: acrylonitrile — General description - α,β -unsaturated nitrile: methacrylonitrile — General description - α,β -unsaturated nitrile: ethacrylonitrile — General description - Unsaturated monocarboxylic/dicarboxylic acids: fumaric acid, maleic acid, acrylic acid, methacrylic acid — General description - Alkyl esters of α,β -unsaturated carboxylic acids: methyl acrylate, ethyl acrylate, butyl acrylate, butyl methacrylate, 2-ethylhexyl acrylate, 2-ethylhexyl methacrylate, octyl acrylate — General description - NBR Mooney viscosity: 24 to 70 ML 1+4 at 100°C — General description - NBR Mooney viscosity: 28 to 40 ML 1+4 at 100°C — General description - NBR Mw: 200,000 to 400,000 — General description - NBR Mw: 200,000 to 500,000 — General description - NBR composition: 40 to 85 wt% diene — Claim 5 - NBR composition: 14 to 50 wt% unsaturated nitrile — Claim 5 - NBR composition: 1 to 10 wt% additional monomers — Claim 5 - Catalyst family: ruthenium or osmium — Claim 1 - Hydrogen present: yes — Claim 1 - Co-olefin: optional — Claim 1

Relevant Experimental Evidence - Example a: the patent reviews prior art metathesis catalysts and explains their limitations for nitrile rubber degradation and hydrogenation. - Example a: the disclosure identifies simultaneous hydrogenation/metathesis as the inventive concept. - Example a: the target product is described as low-RDB HNBR with Mw 20,000 to 250,000 and Mooney 1 to

50. - Example a: the patent points to Hoveyda- and Grela-type Ru complexes as suitable catalyst families. - Example a: the process is positioned as a single-step alternative to sequential metathesis then hydrogenation.

Technical Relevance This patent is highly relevant because it claims the core simultaneous metathesis/hydrogenation route for HNBR synthesis. It is especially important for low-molecular-weight HNBR production and for identifying catalyst families that can operate under hydrogen.

Patent 6

- Patent Number: US7846995B2
- Patent Title: ... -containing catalyst residues from optionally hydrogenated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis itself; instead, it discloses removal of ruthenium-containing catalyst residues from optionally hydrogenated nitrile rubber using functionalized ion-exchange resins. The process is a downstream purification step relevant to HNBR made by ruthenium-catalyzed hydrogenation or metathesis/hydrogenation routes. - Ruthenium content in solution: 5 to 1000 ppm — Claim 2 - Ruthenium content in solution: 5 to 500 ppm — Claim 3 - Rubber concentration in solution: 0.5 to 30 wt% — Claim 4 - Ion exchange resin particle diameter: 0.2 to 2.5 mm dry basis — Claim 1 - Initial Ru content: 15 ppm — Example 2A/2B table - Initial Ru resin sample: 0.5 g — Example 2A - Rubber solution amount: 180 g — Example 2A - Treatment temperature: ca. 100°C — Example 2A - Treatment time: 66 hours — Example 2A - Drying temperature: 60°C — Example 2A - Ru content after treatment: 10 ppm — Example 2A - Ru removal: 33% — Example 2A - Storage stability criterion: not more than 3 Mooney units — Claim 1

Relevant Experimental Evidence - Example 2A: treatment of a nitrile rubber solution with 0.5 g resin at about 100°C for 66 h reduced Ru from 15 ppm to 10 ppm. - Example 2B: the untreated control established the baseline Ru content for comparison. - Example 1: the example block indicates a comparative dataset for catalyst residue removal, though the parser did not extract the full table. - Claim 1: the resin-based purification is intended for optionally hydrogenated nitrile rubber with very low Ru content. - Overall: the patent demonstrates a practical post-treatment route to cleaner HNBR products.

Technical Relevance This patent is relevant because ruthenium-catalyzed HNBR synthesis often leaves residual metal that must be removed for product quality and stability. It provides a concrete purification strategy compatible with hydrogenated or partially hydrogenated nitrile rubber

solutions.

Patent 7

- Patent Number: US7923518B2
- Patent Title: Nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses nitrile rubber precursor manufacture and work-up, not hydrogenation, but it is relevant as a precursor-control patent for HNBR feedstocks. It focuses on emulsion-polymerized NBR with controlled calcium/chlorine content, thio end groups, and storage stability, which can influence subsequent hydrogenation behavior and final HNBR properties. - Calcium content: 150 ppm to 1290 ppm — Claim 1 - Chlorine content: at least 40 ppm — Claim 1 - Storage stability: not more than 3 Mooney units — Claim 1 - Storage stability formula: $MV2 - MV1$ — Claim 1 - Calcium content: 200 ppm to 1290 ppm — Claim 2 - Calcium content: 400 ppm to 1290 ppm — Claim 3 - Calcium content: 500 ppm to 1290 ppm — Claim 4 - Calcium content: 600 ppm to 1290 ppm — Claim 5 - Calcium content: 800 ppm to 1290 ppm — Claim 6 - Acrylonitrile content: 38.6 wt% — Example 36 - Coagulation salt: $CaCl_2$ — Example 36 - Salt solution concentration: 0.3% by weight — Example 36 - Salt based on NBR: 1.8% by weight — Example 36 - Coagulation temperature: 90°C — Example 36 - Washing temperature: 60°C — Example 36 - Washing time: 8 h — Example 36 - Chlorine content: 100 ppm — Example 36 - Mooney viscosity $MV1$: 47 — Example 36 - Mooney viscosity $MV2$: 51 — Example 36 - Storage stability: 4 Mooney units — Example 36

Relevant Experimental Evidence - Example 36: coagulation of latex C with $CaCl_2$ and water containing calcium ions produced NBR with 38.6 wt% acrylonitrile and measurable calcium/chlorine residues. - Example 36: the resulting rubber showed $MV1$ 47 and $MV2$ 51, corresponding to a storage stability of 4 Mooney units. - Example 36: the example illustrates how coagulation and washing conditions affect precursor purity and storage stability. - Claim set: the patent defines a nitrile rubber with controlled calcium/chlorine content and thio end groups. - Overall: the disclosure is a precursor-quality patent rather than an HNBR hydrogenation patent.

Technical Relevance This patent is indirectly relevant because HNBR quality depends strongly on the composition and cleanliness of the NBR precursor. Controlled coagulation, residue levels, and storage stability can affect hydrogenation performance and downstream product consistency.

Patent 8

- Patent Number: US10641418B2

- Patent Title: ... , high-pressure, large-bore bonded flexible rubber pipe and method for producing ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent does not disclose HNBR synthesis; it discloses a bonded flexible pipe structure in which an HNBR-based peroxide-vulcanized rubber compound is bonded to a PVDF liner. The HNBR material is used as a high-performance elastomeric layer, and the patent emphasizes scorch time, peroxide selection, and vulcanization temperature to achieve adhesion. - Internal diameter: 4 inch (102 mm) — Example 1 - Operating pressure: 69 MPa — Example 1 - HNBR layer thickness: 1 mm — Example 1 - Mooney 05, 150°C scorch time: 12.0 min — Example 1 - Peroxide loading: 8 phr — Example 1 - Peroxide active agent: 2,5-dimethyl-2,5-di(t-butylperoxy)-3-hexyne — Example 1 - Peroxide active agent content: 45% — Example 1 - Peroxide half-life: 1 hour at 154°C — Example 1 - Acrylonitrile content: 30% to 40% — Example 1 - Residual double bond content: below 2% — Example 1 - Carbon black loading: 70 phr — Example 1 - Triallyl isocyanate coagent: 2.5 phr — Example 1 - Other additives: 18.7 phr — Example 1 - Vulcanization temperature: 175°C — Example 1 - Adhesion value: above 10 N/mm — Example 1 - Internal diameter: 4 inch (102 mm) — Example 2 - Mooney 05, 150°C scorch time: 5.6 min — Example 2 - Peroxide loading: 8 mass fractions — Example 2 - Peroxide active agent: di(tert-butylperoxy isopropyl)benzene — Example 2 - Peroxide active agent content: 40% — Example 2 - Peroxide half-life: 1 hour at 140°C — Example 2 - Vulcanization temperature: 155°C — Example 2 - Adhesion value: 3.1 N/mm — Example 2 - Internal diameter: 3 inch (76 mm) — Example 3 - Operating pressure: 103.5 MPa — Example 3 - HNBR layer thickness: 1 mm — Example 3 - Mooney 05, 150°C scorch time: 14.0 min — Example 3 - Peroxide loading: 7 phr — Example 3 - Peroxide active agent: di(tert-butylperoxy isopropyl)benzene — Example 3 - Vulcanization temperature: 160°C — Example 3 - Adhesion value: above 15 N/mm — Example 3 - Design pressure: at least 1.5 MPa — General description - Large-bore criterion: internal diameter larger than 25 mm — General description - PVDF copolymer: at least 80 mole% vinylidene fluoride — Claim 1 - Peroxide half-life at 150°C: at least 1 hour — Claim 3 - Scorch-protected peroxide: yes — Claim 5

Relevant Experimental Evidence - Example 1: a 4-inch pipe with 69 MPa operating pressure and HNBR layer cured at 175°C achieved cohesive failure and adhesion above 10 N/mm. - Example 2: lower scorch time and lower vulcanization temperature led to adhesive failure and only 3.1 N/mm adhesion. - Example 3: a 3-inch pipe at 103.5 MPa with 14 min scorch time and 160°C vulcanization achieved cohesive failure and adhesion above 15 N/mm. - Example 1: HNBR with below 2% residual double bond content and 30-40% acrylonitrile was used in the bonded layer. - Overall: scorch-protected peroxide and higher vulcanization temperature improved bonding to PVDF.

Technical Relevance This patent is relevant because it shows a demanding end-use for HNBR and identifies formulation parameters that preserve adhesion and performance in aggressive environments. While not a synthesis patent, it helps define the property window that HNBR synthesis must support, especially low RDB and controlled scorch behavior.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US7666950B2 focuses on direct hydrogenation of NBR in a spinning disk reactor, whereas US7951875B2, US7772328B2, EP2473281B1, and US9150669B2 emphasize metathesis-assisted molecular-weight reduction combined with hydrogenation.
- EP2473281B1 and US9150669B2 are the clearest single-step simultaneous metathesis/hydrogenation disclosures; both target Mw 20,000 to 250,000, Mooney 1 to 50, and low MWD, with hydrogen present and optional co-olefin.
- US7951875B2 and US7772328B2 disclose very similar experimental conditions: 15 wt% or 6 wt% cement, 80°C reactor temperature, 0.05-0.25 phr catalyst loadings, 6.9×10^5 Pa ethylene, and 1-hexene as an alternative co-olefin.
- The metathesis/hydrogenation patents report substantial property reductions, including Mw down to 59,000 and Mooney viscosity down to 3 in the experimental series, showing stronger molecular-weight control than direct hydrogenation-only process claims.
- US7846995B2 is a downstream purification patent rather than a synthesis patent; it addresses Ru residue removal from optionally hydrogenated nitrile rubber using ion-exchange resin, which complements the Ru-catalyzed synthesis routes.
- US7923518B2 is a precursor-quality patent: it controls NBR coagulation residues, calcium/chlorine content, and storage stability, which can influence hydrogenation feedstock quality but does not itself hydrogenate the polymer.
- US10641418B2 is application-driven rather than synthesis-driven; it uses HNBR with below 2% RDB and 30-40% acrylonitrile in peroxide-vulcanized bonded pipe layers, highlighting the property requirements that synthesis must meet.
- Across the set, the most common precursor composition is NBR based on butadiene and acrylonitrile, with optional comonomers such as fumaric acid, maleic acid, acrylic acid, methacrylic acid, and alkyl acrylates/methacrylates.
- Catalyst trends split between homogeneous hydrogenation catalysts for direct HNBR formation and Ru/osmium metathesis-active complexes for tandem molecular-weight reduction plus hydrogenation; post-treatment purification then removes residual Ru.
- Process trends show that lower solids content, controlled temperature, and catalyst loading are used to tune Mw, PDI, and RDB, while downstream applications demand low Mooney, low RDB, and peroxide-curable formulations.

4. CONCLUSION

The evidence shows two dominant synthesis strategies for HNBR: direct selective hydrogenation of NBR, and tandem metathesis/hydrogenation to generate low-molecular-weight HNBR with narrow molecular weight distribution. For direct hydrogenation, the most process-relevant disclosure is the spinning disk reactor approach with hydrogen pressures from 3.5 to 100 bar, which is intended to reduce mass-transfer limitations and equipment cost. For low-Mooney HNBR, the strongest evidence supports simultaneous metathesis/hydrogenation in the presence of hydrogen and Ru- or osmium-based catalysts, with target products in the Mw 20,000 to 250,000 range, Mooney 1 to 50, and low residual double bond content. Experimental data indicate that catalyst loading, solids content, co-olefin choice, and temperature materially affect Mw, PDI, and RDB. Recommended synthesis parameters based on the evidence are: use butadiene/acrylonitrile-based NBR precursors, maintain controlled precursor purity and storage stability, apply Ru-based simultaneous metathesis/hydrogenation when low molecular weight is desired, and follow with ruthenium residue removal by ion exchange when necessary. For end-use performance, HNBR with acrylonitrile content around 30-40% and residual double bond content below 2% is consistent with demanding oilfield and flexible-pipe applications.

5. REFERENCES

- US7666950B2 | Process for preparing hydrogenated nitrile rubbers | Not disclosed | US | 2010 | <https://patents.google.com/patent/US7666950B2/en>
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